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A magnetic moment is equivalent to a current loop!

$$\vec{\mu} = I \vec{A}$$



Suppose an electron of charge $-e$ moves in a circular loop of radius R . If T is the time taken to complete one revolution,

$$I = \frac{e}{T} \Rightarrow |\vec{\mu}| = \frac{e}{T} \pi R^2 v$$

Now $T = \frac{2\pi R}{v}$, v is the speed of the electron.

$$\Rightarrow |\vec{\mu}| = \frac{e v}{2\pi R} \cdot \pi R^2 = \frac{1}{2} e v R$$

If m is the mass of the electron, the angular momentum $\vec{L}$ is

$$|\vec{L}| = m v R, \text{ perpendicular to the plane of the loop}$$

$$\Rightarrow |\vec{\mu}| = \frac{e}{2m} \cdot m v R = \frac{e}{2m} |\vec{L}|$$

In vector notation, $\boxed{\vec{\mu} = -\frac{e}{2m} \vec{L}}$

In classical mechanics, only orbital motion gives rise to magnetic moment. However, in Q Mech, in addition to the magnetic moment due to the orbital motion, the electron has an "intrinsic" magnetic moment which is attributed to the "spin" angular momentum of electron, denoted by $\vec{S}$. Moreover, we multiply the right hand side with a constant called the g-factor 'g', such that

$$|\vec{\mu}| = g \cdot \frac{e}{2m} |\vec{S}| = g \cdot \frac{e\hbar}{2m} \left(\frac{|\vec{S}|}{\hbar} \right) \rightarrow \text{dimensionless}$$

dimensionless, 2 for electron

μ_B called Bohr magneton

$$\mu_B = 9.27 \times 10^{-24} \text{ J/T}$$

So, $\vec{\mu} = -g \mu_B \frac{\vec{S}}{\hbar}$, where $g=2$

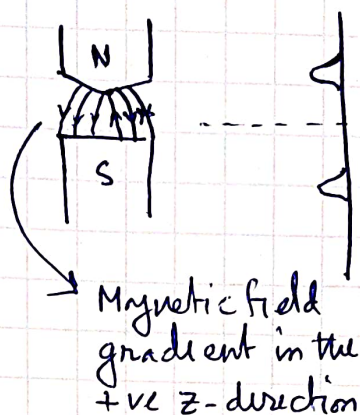
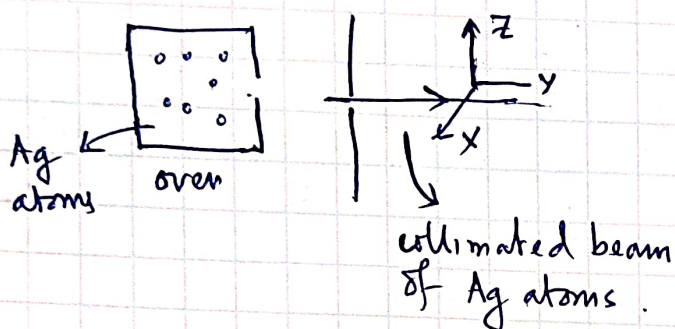
$$\mu_B = \frac{e\hbar}{2m}$$

since electron charge is negative $\vec{\mu}$ and $\vec{S}$ are antiparallel.

Stern-Gerlach experiment

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Force on a dipole moment $\vec{\mu}$ placed in a magnetic field $\vec{B}$ is given by, $\vec{F} = \nabla(\vec{\mu} \cdot \vec{B})$, in the direction $\vec{\mu} \cdot \vec{B}$ increases the fastest.



Ag atom has 47 electrons. 46 electrons completely fill the $n=1, 2, 3, 4$ levels. The remaining electron is in $5s$ state ($n=5, l=0$) $\Rightarrow$ no orbital angular momentum.

$$\vec{F} = \nabla(\vec{\mu} \cdot \vec{B}) = \nabla(\mu_z B_z) = \mu_z \frac{dB_z}{dz} \hat{z}$$

Magnetic moments of incident Ag atoms ~~are~~ are randomly oriented. Classically, one should expect the moments to be deflected along z -direction by a random amount $\Rightarrow$ a continuous, smooth distribution of Ag atoms detected by the screen.

Observed result: two separate peaks $\rightarrow$ two separate strips of Ag atoms on the screen.

$\rightarrow$ quantization of the magnetic dipole moment.

$\rightarrow$ quantization of spin angular momentum.

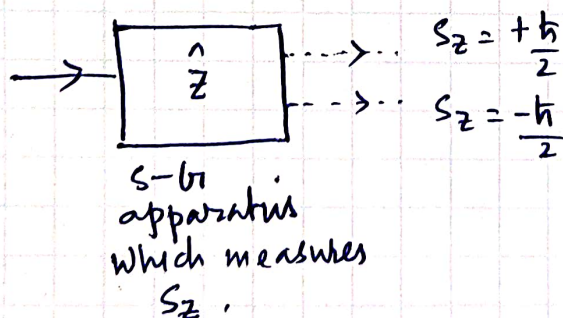
$$\mu_z = -2\mu_B \frac{S_z}{\hbar}$$

It turns out, $\frac{S_z}{\hbar} = \pm \frac{1}{2}$

$$\Rightarrow \boxed{S_z = \pm \frac{\hbar}{2}}$$

$\rightarrow$ spin-half particles.

$\mu_B \rightarrow$ magnitude of electron's magnetic moment.

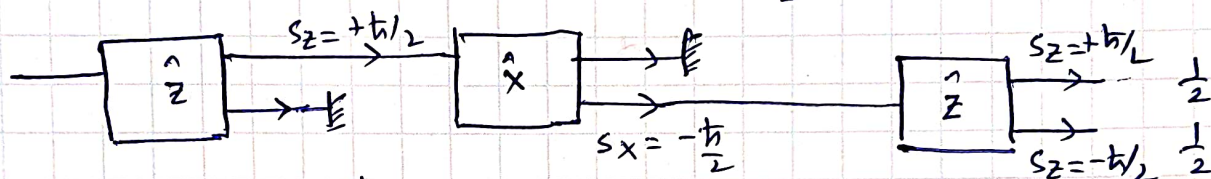
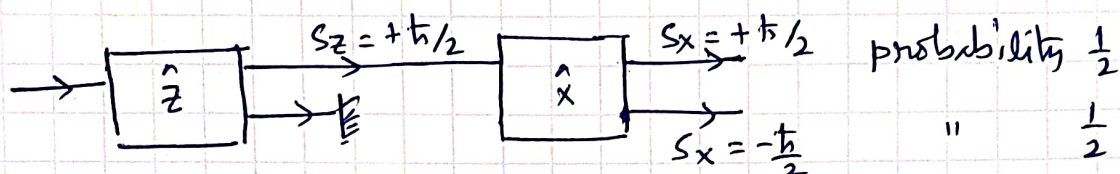
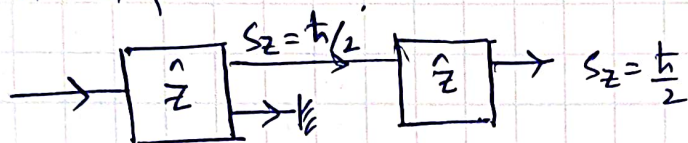


The spin state of the electron can be described using two basis vectors ^③
 $|z; +\rangle$, $|z; -\rangle$, such that,

$$\hat{S}_z |z; +\rangle = +\frac{\hbar}{2} |z; +\rangle$$

$$\hat{S}_z |z; -\rangle = -\frac{\hbar}{2} |z; -\rangle$$

Some more experiments!



A purely quantum mechanical effect, ~~classical~~ no classical analogue.

$|z; \pm\rangle$ can be written as a superposition of $|x; \pm\rangle$ states and vice versa! A state with definite value of S_z has an amplitude along the states corresponding to $S_x = \pm\hbar/2$

Concrete representations of $|z; \pm\rangle$ as column vectors:

$$\begin{aligned} |z; +\rangle &= |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |z; -\rangle &= |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned}$$

two basis states in a ~~2D~~ 2-dimensional complex vector space.

Any general vector in the 2D complex vector space can be expressed as,

$$|\psi\rangle = c_1 |\uparrow\rangle + c_2 |\downarrow\rangle$$

c_1, c_2 are complex numbers.

$$= c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

Spin- $\frac{1}{2}$ particle : two-state system

(4)

Define "inner products" : * $(\phi, \psi) = \int dx \phi^*(x) \psi(x)$

* If $|\psi_1\rangle = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix}$, $|\psi_2\rangle = \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix}$

$$\langle \psi_1, \psi_2 \rangle = \alpha_1^* \alpha_2 + \beta_1^* \beta_2 = (\alpha_1^* \ \beta_1^*) \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} = \langle \psi_1 | \psi_2 \rangle$$

Bra $\swarrow$ ket

If an operator $\hat{H}$ is hermitian, $(\hat{H}^T)^* = \hat{H}$

A general 2×2 Hermitian matrix

$$\hat{H} = \begin{pmatrix} a_0 + a_3 & a_1 - ia_2 \\ a_1 + ia_2 & a_0 - a_3 \end{pmatrix}$$

$$= a_0 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + a_1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + a_3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$
$$= a_0 \sigma_0 + a_1 \sigma_1 + a_2 \sigma_2 + a_3 \sigma_3 = \sum_{i=1}^4 a_i \sigma_i$$

We can write down a generic 2×2 Hamiltonian

$$\hat{H} = \frac{\hbar \omega_1}{2} \sigma_1 + \frac{\hbar \omega_2}{2} \sigma_2 + \frac{\hbar \omega_3}{2} \sigma_3 \quad [\sigma_1, \sigma_2, \sigma_3 \rightarrow \text{Pauli matrices}]$$

(σ_0 adds only a constant term to the Hamiltonian and hence not physically interesting)

$$\Rightarrow \hat{H} = \omega_1 \left(\frac{\hbar}{2} \sigma_1 \right) + \omega_2 \left(\frac{\hbar}{2} \sigma_2 \right) + \omega_3 \left(\frac{\hbar}{2} \sigma_3 \right)$$
$$= \omega_1 \hat{S}_x + \omega_2 \hat{S}_y + \omega_3 \hat{S}_z$$

where, $\hat{S}_x = \frac{\hbar}{2} \sigma_1 = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$\hat{S}_y = \frac{\hbar}{2} \sigma_2 = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{S}_z = \frac{\hbar}{2} \sigma_3 = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The Pauli matrices $\sigma_1, \sigma_2, \sigma_3$ along with the identity matrix $\sigma_0 = \mathbb{I}$ form a complete set of 2×2 matrices that can be viewed as basis vectors in a that can be used to build the most general Hermitian 2×2 matrix.

Let us verify that the matrix representation of $\hat{S}_z$ satisfies (5)

$$\hat{S}_z |z; \pm\rangle = \pm \frac{\hbar}{2} |z; \pm\rangle$$

$$\hat{S}_z |z; +\rangle = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = +\frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = +\frac{\hbar}{2} |z; +\rangle$$

$$\hat{S}_z |z; -\rangle = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{\hbar}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{\hbar}{2} |z; -\rangle$$

$|z; \pm\rangle$ are as ~~are~~ represented by column vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are the eigenstates of matrix $\hat{S}_z$.

Spin is intrinsic angular momentum. We already know the commutation relations of the orbital angular momentum components.

$$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y$$

While L_i 's are hermitian operators expressed in terms of co-ordinates and momenta, $\hat{S}_i$'s are more abstract: they act on column vectors rather than on wave functions. However, if spin is also an angular momentum operator, it must obey the same commutation relations as $\hat{L}_x, \hat{L}_y, \hat{L}_z$.

$$\begin{aligned} [\hat{S}_x, \hat{S}_y] &= \left[\frac{\hbar}{2} \sigma_1, \frac{\hbar}{2} \sigma_2 \right] = \left(\frac{\hbar}{2} \right)^2 (\sigma_1 \sigma_2 - \sigma_2 \sigma_1) \\ &= \left(\frac{\hbar}{2} \right)^2 \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] \\ &= \left(\frac{\hbar}{2} \right)^2 \left[\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \right] \\ &= \left(\frac{\hbar}{2} \right)^2 \left[\begin{pmatrix} 2i & 0 \\ 0 & -2i \end{pmatrix} \right] = \frac{\hbar^2}{2} i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= i\hbar \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = i\hbar \hat{S}_z \end{aligned}$$

$$\Rightarrow [\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z$$

$$\text{Similarly, } [\hat{S}_y, \hat{S}_z] = i\hbar \hat{S}_x$$

$$[\hat{S}_z, \hat{S}_x] = i\hbar \hat{S}_y$$

$$\text{In compact notation, } [\hat{S}_i, \hat{S}_j] = i\hbar \epsilon_{ijk} \hat{S}_k$$

Column vector representations of the eigenstates of $\hat{S}_x, \hat{S}_y$

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Let us try, $\hat{S}_x \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{\hbar}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{\hbar}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Again, $\hat{S}_x \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{\hbar}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{\hbar}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -\frac{\hbar}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

We write $|x; +\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} [| \uparrow \rangle + | \downarrow \rangle]$

and $|x; -\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} [| \uparrow \rangle - | \downarrow \rangle]$

Also $\hat{S}_x |x; \pm\rangle = \pm \frac{\hbar}{2} |x; \pm\rangle$

Finally, $\hat{S}_y \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{\hbar}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$

Again, $\hat{S}_y \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{\hbar}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ i \end{pmatrix} = -\frac{\hbar}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$

We write, $|y; +\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ i \end{pmatrix} \right] = \frac{1}{\sqrt{2}} [| \uparrow \rangle + i | \downarrow \rangle]$

$|y; -\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ i \end{pmatrix} \right] = \frac{1}{\sqrt{2}} [| \uparrow \rangle - i | \downarrow \rangle]$

Also, $\hat{S}_y |y; \pm\rangle = \pm \frac{\hbar}{2} |y; \pm\rangle$

To summarize, $\hat{S}_i = \frac{\hbar}{2} \sigma_i$, where $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$\hat{S}_z |z; +\rangle = \hat{S}_z | \uparrow \rangle = +\frac{\hbar}{2} | \uparrow \rangle$, $\hat{S}_z |z; -\rangle = \hat{S}_z | \downarrow \rangle = -\frac{\hbar}{2} | \downarrow \rangle$

$|x; +\rangle = \frac{1}{\sqrt{2}} [| \uparrow \rangle + | \downarrow \rangle] = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 $|x; -\rangle = \frac{1}{\sqrt{2}} [| \uparrow \rangle - | \downarrow \rangle] = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ } $\hat{S}_x |x; \pm\rangle = \pm \frac{\hbar}{2} |x; \pm\rangle$

$|y; +\rangle = \frac{1}{\sqrt{2}} [| \uparrow \rangle + i | \downarrow \rangle] = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$
 $|y; -\rangle = \frac{1}{\sqrt{2}} [| \uparrow \rangle - i | \downarrow \rangle] = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$ } $\hat{S}_y |y; \pm\rangle = \pm \frac{\hbar}{2} |y; \pm\rangle$

Now verify yourself that these relations are consistent with the thought experiments discussed before.

Also, $[\hat{S}_i, \hat{S}_j] = i\hbar \epsilon_{ijk} \hat{S}_k \Rightarrow [\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k$

Spin precession: time evolution of spin $\frac{1}{2}$ particle under uniform magnetic field. (10)

$$\hat{H} = -\vec{\mu} \cdot \vec{B} = -\frac{e}{m} \vec{S} \cdot \vec{B}$$

If $\vec{B} = B \hat{z}$, $\hat{H} = -\frac{eB}{m} S_z = -\frac{e\hbar}{2m} B \sigma_z = -\mu_B B \sigma_z$

$[\hat{S}_z, \hat{H}] = 0 \Rightarrow \hat{S}_z, \hat{H}$ have simultaneous eigenstates.

$$\left. \begin{aligned} \hat{H}|\uparrow\rangle &= E_{\uparrow}|\uparrow\rangle \\ \hat{H}|\downarrow\rangle &= E_{\downarrow}|\downarrow\rangle \end{aligned} \right\} E_{\uparrow} = -\mu_B B, E_{\downarrow} = \mu_B B$$

We define $|E_{\uparrow} - E_{\downarrow}| = \hbar\omega \Rightarrow \hbar\omega = 2\mu_B B$

$$\Rightarrow \omega = \frac{e\hbar B}{m} \rightarrow \text{cyclotron frequency}$$

$$\hat{H} = -\omega S_z = -\frac{\hbar}{2} \omega \sigma_z$$

Time evolution operator $U(t, 0) = \exp\left(-\frac{i\hat{H}t}{\hbar}\right) = \exp\left(+\frac{i\omega t}{2} \sigma_z\right)$

Suppose, at $t=0$, $|\psi(0)\rangle = a_{\uparrow}|\uparrow\rangle + b_{\downarrow}|\downarrow\rangle$

$$|\psi(t)\rangle = a_{\uparrow} \exp\left(\frac{i\omega t}{2}\right) |\uparrow\rangle + b_{\downarrow} \exp\left(-\frac{i\omega t}{2}\right) |\downarrow\rangle$$

[Here $\sigma_z|\uparrow\rangle = |\uparrow\rangle$, $\sigma_z|\downarrow\rangle = -|\downarrow\rangle$]

Case: 1 Initial state $|\psi(0)\rangle = |\uparrow\rangle \Rightarrow a_{\uparrow}=1, b_{\downarrow}=0$ at $t=0$

$$|\psi(t)\rangle = \exp\left(+\frac{i\omega t}{2}\right) |\uparrow\rangle$$

System remains in the spin-up state, stationary state.

Case: 2 At $t=0$, the particle is in the $|S_x, +\rangle$ state.

$$|\psi(0)\rangle = |S_x, +\rangle = \frac{1}{\sqrt{2}} [|\uparrow\rangle + |\downarrow\rangle], a_{\uparrow} = b_{\downarrow} = \frac{1}{\sqrt{2}}$$

Find the probability for the particle to be in the $|S_x, +\rangle$ state at some later time t .

⑪

$$\begin{aligned}
 S_{x,t} &= |\langle S_x \rangle + |\psi(t)\rangle|^2 = \left| \left(\frac{1}{\sqrt{2}} \langle \uparrow | + \frac{1}{\sqrt{2}} \langle \downarrow | \right) \left[\frac{1}{\sqrt{2}} \exp\left(\frac{i\omega t}{2}\right) |\uparrow\rangle + \frac{1}{\sqrt{2}} \exp\left(-\frac{i\omega t}{2}\right) |\downarrow\rangle \right] \right|^2 \\
 &= \left| \frac{1}{2} \exp\left(\frac{i\omega t}{2}\right) + \frac{1}{2} \exp\left(-\frac{i\omega t}{2}\right) \right|^2 \\
 &= \cos^2 \frac{\omega t}{2}
 \end{aligned}$$

Similarly, $P_{S_x,-} = \sin^2 \frac{\omega t}{2}$

$$\Rightarrow \langle S_x \rangle = P_{S_x,+} \left(\frac{+\hbar}{2}\right) + P_{S_x,-} \left(-\frac{\hbar}{2}\right) = \frac{\hbar}{2} \cos \omega t$$

Similarly $\langle S_y \rangle = \frac{\hbar}{2} \sin \omega t$

$$\langle S_z \rangle = 0$$

Magnetic field in the z-direction: spin-precession in the xy plane.

ex 3 $B = B \hat{x}$, initial state $|\psi(0)\rangle = |\uparrow\rangle$

$$\begin{aligned}
 U(t,0) &= \exp\left(-\frac{i\omega t}{2} \sigma_x\right) \quad \left[\because \hat{H} = -\frac{eB}{m} S_x = -\mu_B B \sigma_x \right] \\
 &\quad \left[\Rightarrow \hat{H} = -\frac{\hbar}{2} \omega \sigma_x \right]
 \end{aligned}$$

Use formula: $\exp(-iA \hat{n} \cdot \vec{\sigma}) = \cos A \mathbb{I} - i(\hat{n} \cdot \vec{\sigma}) \sin A$

Here $A \hat{n} = \frac{\omega t}{2} \hat{x} \Rightarrow \exp\left(-\frac{i\omega t}{2} \sigma_x\right) = \cos \frac{\omega t}{2} \mathbb{I} - i \sigma_x \sin \frac{\omega t}{2}$

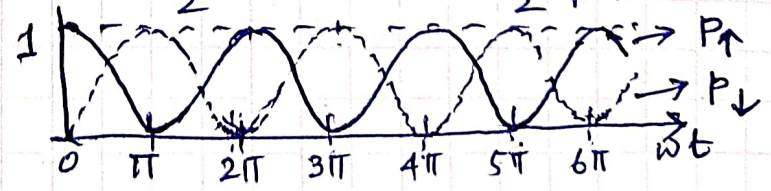
where $\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$; $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$|\psi(t)\rangle = \exp\left(-\frac{i\omega t}{2} \sigma_x\right) |\uparrow\rangle = \begin{pmatrix} \cos \frac{\omega t}{2} & -i \sin \frac{\omega t}{2} \\ -i \sin \frac{\omega t}{2} & \cos \frac{\omega t}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

probabilities for the particle to be in the $|\uparrow\rangle, |\downarrow\rangle$ states

$$P_{\uparrow} = \cos^2 \frac{\omega t}{2}, \quad P_{\downarrow} = \sin^2 \frac{\omega t}{2}$$

$$\begin{aligned}
 &= \begin{pmatrix} \cos \frac{\omega t}{2} \\ -i \sin \frac{\omega t}{2} \end{pmatrix} = \cos \frac{\omega t}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} - i \sin \frac{\omega t}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
 &= \cos \frac{\omega t}{2} |\uparrow\rangle - i \sin \frac{\omega t}{2} |\downarrow\rangle
 \end{aligned}$$



Spin states in arbitrary direction

(12)

consider the unit vector $\hat{n} = (n_x, n_y, n_z)$

$$\vec{S} = (\hat{S}_x, \hat{S}_y, \hat{S}_z) = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$$

such that $\hat{S} = \frac{\hbar}{2} \hat{\sigma}$

Define $\hat{S}_n = \hat{n} \cdot \hat{S} = n_x \hat{S}_x + n_y \hat{S}_y + n_z \hat{S}_z = \frac{\hbar}{2} \hat{n} \cdot \hat{\sigma}$

The eigenvalues of $\hat{S}_n$ are $\pm \frac{\hbar}{2}$.

$$(\hat{S}_n)^2 = \left(\frac{\hbar}{2}\right)^2 (\hat{n} \cdot \hat{\sigma})^2 = \left(\frac{\hbar}{2}\right)^2$$

$$\begin{aligned}\hat{S}_n &= \frac{\hbar}{2} \left[n_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + n_y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + n_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \\ &= \frac{\hbar}{2} \begin{bmatrix} n_z & n_x - i n_y \\ n_x + i n_y & -n_z \end{bmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta \end{pmatrix}\end{aligned}$$

$$\hat{S}_n |\vec{n}; \pm\rangle = \pm \frac{\hbar}{2} |\vec{n}; \pm\rangle \quad \left[\begin{array}{l} \text{eigenvalues are} \\ \text{roots of } \det |\hat{S}_n - \lambda \mathbb{I}| = 0 \end{array} \right]$$

$|\vec{n}; \pm\rangle$ represents a spin state that points 'up' along $\vec{n}$ and a spin state that points 'down' along $\vec{n}$, respectively.

To find the representation of $|\vec{n}; \pm\rangle$ we write

$$|\vec{n}; \pm\rangle = c_1 |\uparrow\rangle + c_2 |\downarrow\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

The eigenvalue equation: $(\hat{S}_n - \frac{\hbar}{2} \mathbb{I}) |\vec{n}; \pm\rangle = 0$

$$\frac{\hbar}{2} \begin{pmatrix} \cos\theta - 1 & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta - 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0$$

$$\Rightarrow c_2 = e^{i\phi} \frac{1 - \cos\theta}{\sin\theta} c_1 = e^{i\phi} \frac{\sin\theta/2}{\cos\theta/2} c_1$$

$$\text{Again } |c_1|^2 + |c_2|^2 = 1 \Rightarrow |c_1|^2 \left[1 + \frac{\sin^2\theta/2}{\cos^2\theta/2} \right] = 1$$

$$\Rightarrow |c_1|^2 = \cos^2\theta/2$$

$$\text{we choose, } c_1 = \cos\theta/2, \quad c_2 = \sin\theta/2 e^{i\phi}$$

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$$\Rightarrow |\vec{n}; +\rangle = \cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} e^{2i\varphi} |\downarrow\rangle$$

Similarly, we get,

$$|\vec{n}; -\rangle = -\sin \frac{\theta}{2} |\uparrow\rangle + \cos \frac{\theta}{2} e^{2i\varphi} |\downarrow\rangle$$

often, It is convenient to multiply the above expression by $+e^{-2i\varphi}$

Final representations of the $|\vec{n}; \pm\rangle$ eigenstates:

$$\begin{aligned} |\vec{n}; +\rangle &= \cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} e^{2i\varphi} |\downarrow\rangle \\ |\vec{n}; -\rangle &= -\sin \frac{\theta}{2} e^{-2i\varphi} |\uparrow\rangle + \cos \frac{\theta}{2} |\downarrow\rangle \end{aligned}$$

Verify that $|\vec{n}; \pm\rangle$ reduces to the known results along z, x, y directions.

Matrix representation: $|\vec{n}; +\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{2i\varphi} \end{pmatrix}$

$$|\vec{n}; -\rangle = \begin{pmatrix} -\sin \frac{\theta}{2} \\ \cos \frac{\theta}{2} e^{2i\varphi} \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} -\sin \frac{\theta}{2} e^{-2i\varphi} \\ \cos \frac{\theta}{2} \end{pmatrix}$$

Verify that $\langle \vec{n}; + | \vec{n}; - \rangle = 0$

and $\langle \vec{n}; \pm | \vec{n}; \pm \rangle = 1$

consider a spin $\frac{1}{2}$ particle to be in the state $|\psi(0)\rangle = |z; +\rangle = |\uparrow\rangle$ at $t=0$. Now a magnetic field B is applied in the $\hat{n}$ direction calculate the probability of finding the particle in $|\downarrow\rangle$ state at time t .

Ans $\hat{H} = -\omega \hat{S}_{\hat{n}}$, $|\psi(0)\rangle = |\uparrow\rangle$

$$|\hat{n}; +\rangle = \cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} e^{2\varphi} |\downarrow\rangle \quad \dots \textcircled{1}$$

$$|\hat{n}; -\rangle = -\sin \frac{\theta}{2} e^{-2\varphi} |\uparrow\rangle + \cos \frac{\theta}{2} |\downarrow\rangle$$

$$\textcircled{1} \times \cos \frac{\theta}{2} \quad \quad \quad \equiv -\sin \frac{\theta}{2} |\uparrow\rangle + \cos \frac{\theta}{2} e^{2\varphi} |\downarrow\rangle \quad \dots \textcircled{2}$$

$$-\textcircled{2} \times \sin \frac{\theta}{2} \Rightarrow \boxed{|\uparrow\rangle = \cos \frac{\theta}{2} |\hat{n}; +\rangle - \sin \frac{\theta}{2} |\hat{n}; -\rangle}$$

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\uparrow\rangle$$

$$= e^{i\omega \hat{S}_{\hat{n}} t/\hbar} \left[\cos \frac{\theta}{2} |\hat{n}; +\rangle - \sin \frac{\theta}{2} |\hat{n}; -\rangle \right]$$

$$= \cos \frac{\theta}{2} e^{i\omega t/2} |\hat{n}; +\rangle - \sin \frac{\theta}{2} e^{-i\omega t/2} |\hat{n}; -\rangle$$

$$\langle \downarrow | \psi(t) \rangle = \cos \frac{\theta}{2} e^{2i\omega t/2} \langle \downarrow | \hat{n}; +\rangle - \sin \frac{\theta}{2} e^{-i\omega t/2} \langle \downarrow | \hat{n}; -\rangle$$

Here, $\langle \downarrow | \hat{n}; +\rangle = \sin \frac{\theta}{2} e^{2\varphi}$, $\langle \downarrow | \hat{n}; -\rangle = \cos \frac{\theta}{2} e^{2\varphi}$

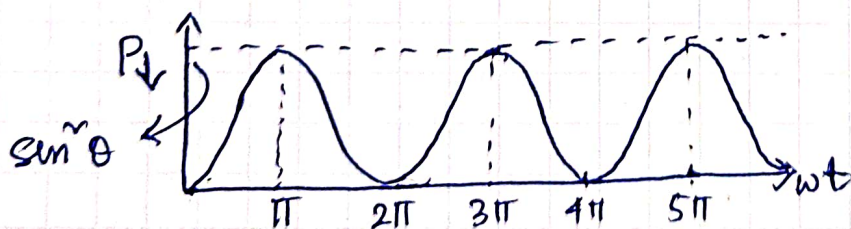
$$\Rightarrow \langle \downarrow | \psi(t) \rangle = \cos \frac{\theta}{2} e^{2i\omega t/2} \sin \frac{\theta}{2} e^{2\varphi} - \sin \frac{\theta}{2} e^{-i\omega t/2} \cos \frac{\theta}{2} e^{2\varphi}$$

$$= i \sin \theta e^{2\varphi} \left(\frac{e^{2i\omega t/2} - e^{-i\omega t/2}}{2i} \right)$$

$$= i \sin \theta e^{2\varphi} \sin \omega t/2$$

Probability of finding the particle in $|\downarrow\rangle$ is: $|\langle \downarrow | \psi(t) \rangle|^2$

$$\Rightarrow |\langle \downarrow | \psi(t) \rangle|^2 = \sin^2 \theta \sin^2 \frac{\omega t}{2}, \quad \text{where } \omega = \frac{eB}{m}$$



$$(P_{\downarrow})_{\max} = \sin^2 \theta$$